ACTIVIDAD:

IPSec VPN

Escuela:

Universidad Politécnica de San Luis Potosi

Materia:

CNO V – Seguridad Informática

Estudiante:

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Introducción.

En esta actividad se realizó en la implementación de una red utilizando una VPN con el protocolo IPSec. El objetivo principal fue conectar dos redes locales diferentes a través de una red pública, la cual fue simulada por el router ISP de manera que la información viaje encriptada y segura. Para esto, se configuraron dos routers (R1 y R2) para crear un túnel por donde pasarán los datos de una PC a otra, asegurando que si alguien intercepta el tráfico en medio, no pueda leerlo a menos que conozca la llave privada.

Desarrollo.

1. Configuración Inicial.

Durante la configuración inicial se configuraron los routers, las interfaces G0/0 y G0/1 dando direcciones IP. Se agregaron IPs estáticas las cuales apuntaban al ISP para poder realizar correctamente la transferencia de datos. Además, se asignaron IPs a las 2 PC utilizadas.

Configuración de IP en router ISP:

```
Router>EN
Router>ENable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface G0/0
Router(config-if)#hostname ISP
ISP(config)#ip address 209.165.100.2 255.255.255.0
                ^
% Invalid input detected at '^' marker.

ISP(config)#interface G0/0
ISP(config-if)#ip address 209.165.100.2 255.255.255.0
ISP(config-if)#no shut

ISP(config-if)#no shut
ISP(config-if)#interface G0/1
ISP(config-if)#ip address 209.165.200.2 255.255.255.0
ISP(config-if)#no shut

ISP(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.1
ISP(config)#exit
ISP#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
ISP#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340
```

Configuración router R1:

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#interface G0/0
R1(config-if)#IP ADDress 209.165.100.1 255.255.255.0
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.100.2
```

Configuración router R2:

```
Router>en
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#int
R2(config)#interface G0/1
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#interface G0/0
R2(config-if)#ip address 209.165.200.1 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.2
R2(config)#
```

2. Licencia de seguridad habilitada.

Se utilizo el comando: **license boot module c1900 technology-package securityk9.** Esto con el fin de activar el paquete de seguridad que permite el uso de comandos de criptografía, posteriormente se reiniciaron los router R1 y R2 para su correcta implementación.

Router 1:

```
R1(config)#license boot module c1900 technology-package securityk9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
together with an additional  payment.  You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the  product,  including  during the 60 day  evaluation  period,  is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day  evaluation  period,  your  use of the  product  feature will be
governed  solely by the Cisco  end user license agreement (link above),
together  with any supplements  relating to such product  feature.  The
above  applies  even if the evaluation  license  is  not  automatically
terminated  and you do  not receive any notice of the expiration of the
evaluation  period.  It is your  responsibility  to  determine when the
evaluation  period is complete and you are required to make  payment to
Cisco for your use of the product feature beyond the evaluation period.

Your  acceptance  of  this agreement  for the software  features on one
product  shall be deemed  your  acceptance  with  respect  to all  such
software  on all Cisco  products  you purchase  which includes the same
software.  (The foregoing  notwithstanding, you must purchase a license
for each software  feature you use past the 60 days evaluation  period,
so  that  if you enable a software  feature on  1000  devices, you must
purchase 1000 licenses for use past  the 60 day evaluation period.)

Activation  of the  software command line interface will be evidence of
your acceptance of this agreement.
```

```
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
R1#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
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Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
########################################################################## [OK]
Smart Init is enabled
smart init is sizing iomem
                  TYPE      MEMORY_REQ
     Onboard devices &
          buffer pools      0x01E8F000
-----------------------------------------------
                TOTAL:      0x01E8F000
Rounded IOMEM up to: 32Mb.
Using 6 percent iomem. [32Mb/512Mb]

              Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
```

Router 2:

```
R2(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.2
R2(config)#license boot module c1900 technology-package securityk9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO  BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires  an additional license from Cisco,
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on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the  product,  including  during the 60 day  evaluation  period,  is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day  evaluation  period,  your  use of the  product  feature will be
governed  solely by the Cisco  end user license agreement (link above),
together  with any supplements  relating to such product  feature.  The
above  applies  even if the evaluation  license  is  not  automatically
terminated  and you do  not receive any notice of the expiration of the
evaluation  period.  It is your  responsibility  to  determine when the
evaluation  period is complete and you are required to make  payment to
Cisco for your use of the product feature beyond the evaluation period.

Your  acceptance  of  this agreement  for the software  features on one
product  shall be deemed  your  acceptance  with  respect  to all  such
software  on all Cisco  products  you purchase  which includes the same
software.  (The foregoing  notwithstanding, you must purchase a license
for each software  feature you use past the 60 days evaluation  period,
so  that  if you enable a software  feature on  1000  devices, you must
purchase 1000 licenses for use past  the 60 day evaluation period.)

Activation  of the  software command line interface will be evidence of
your acceptance of this agreement.


ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot

R2(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module name = C1900 Next reboot level
= securityk9 and License = securityk9

R2(config)#
```

3. Implementación de ACLs.

Aquí se utilizo el comando **Access-list 100 permit ip …..**

Este comando es muy importante ya que si la IP coincide lo que hará será atrapar el mensaje y mandarlo a través de la VPN con el fin de asegurarlo y dificultar que otras personas puedan interceptarlo.

Router 1:

```
R1(config-if)#exit
R1(config)#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

Router 2:

```
R2>
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#access-list 100 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
R2(config)#
```

4. ISAKMP policy.

encryption aes 256: Se utiliza para cifrar los datos de forma robusta.

authentication pre-share: Da el permiso a poder usar una llave compartida.

group 5: Intercambio de claves Diffie-Hellman.

crypto isakmp key: Se coloco la contraseña ("secretkey") y la dirección IP del router vecino para que se puedan reconocer al tener esta misma llave.

Router 1:

```
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#
R1(config)#cry
R1(config)#crypto isakmp key secretkey address 209.165.200.1
```

Router 2:

```
R2(config)#crypto isakmp policy 10
R2(config-isakmp)#encryption aes 256
R2(config-isakmp)#authentication pre-share
R2(config-isakmp)#group 5
R2(config-isakmp)#
R2(config-isakmp)#exit
R2(config)#
R2(config)#crypto isakmp key secretkey address 209.165.100.1
```

5. IPSec transform-set.

Se define como se encriptarán los datos. Se utiliza **crypto ipsec transform-set** definiendo **esp-aes 256** para el cifrado de los datos y **esp-sha-hmac** para la integridad de estos mismos.

Router 1:

```
R1(config)#crypto ipsec transform-set R1->R2 esp-aes 256 esp-sha-hmac
R1(config)#
```

Router 2:

```
R2(config)#crypto ipsec transform-set R2->R1 esp-aes 256 esp-sha-hmac
R2(config)#
```

6. Crear el mapa criptográfico.

Aquí se junta todo lo anterior para crear un mapa en el cual solo las personas identificadas y con acceso podrán recibir los datos de la otra computadora. Se define el destino con peer, se asignan las reglas de encriptación (transform-set) y se identifica el trafico especifico utilizando match address 100.

Router 1:

```
R1(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#set peer 209.165.200.1
R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)#
R1(config-crypto-map)#exit
```

Router 2:

```
R2(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R2(config-crypto-map)#
R2(config-crypto-map)#set peer 209.165.100.1
R2(config-crypto-map)#set pfs group5
R2(config-crypto-map)#set security-association lifetime seconds 86400
R2(config-crypto-map)#set transform-set R2->R1
R2(config-crypto-map)#match address 100
R2(config-crypto-map)#exit
```

7. Aplicar el mapa criptográfico.

Se activan las interfaces de salida, todo lo que salga por G0/0 se revisara y si coincide se encriptara y se mandara hacia el otro lado.

Router 1:

```
R1(config)#int g0/0
R1(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#
```

Router 2:

```
R2(config)#int g0/0
R2(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R2(config-if)#
R2(config-if)#
```

8. Prueba de recibimiento de paquetes: PC-A PC-B